

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

Practical – 3

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AIM: Finding Relationships Among Variables:

- A. Use a dataset with multiple variables and perform correlation analysis to determine the strength and direction of relationships between pairs of variables.**

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load Dataset

df = pd.read_csv("housing.csv")

# Display First Five Records

print(df.head())

# Display Numerical Columns

print(df.select_dtypes(include="number").columns)
```

```
***
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms \
0    -122.23    37.88                41.0         880.0         129.0
1    -122.22    37.86                21.0        7099.0        1106.0
2    -122.24    37.85                52.0        1467.0         190.0
3    -122.25    37.85                52.0        1274.0         235.0
4    -122.25    37.85                52.0        1627.0         280.0

   population  households  median_income  median_house_value  ocean_proximity
0         322.0        126.0          8.3252         452600.0      NEAR BAY
1        2401.0       1138.0          8.3014        358500.0      NEAR BAY
2         496.0        177.0          7.2574        352100.0      NEAR BAY
3         558.0        219.0          5.6431        341300.0      NEAR BAY
4         565.0        259.0          3.8462        342200.0      NEAR BAY
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
      'total_bedrooms', 'population', 'households', 'median_income',
      'median_house_value'],
      dtype='object')
```

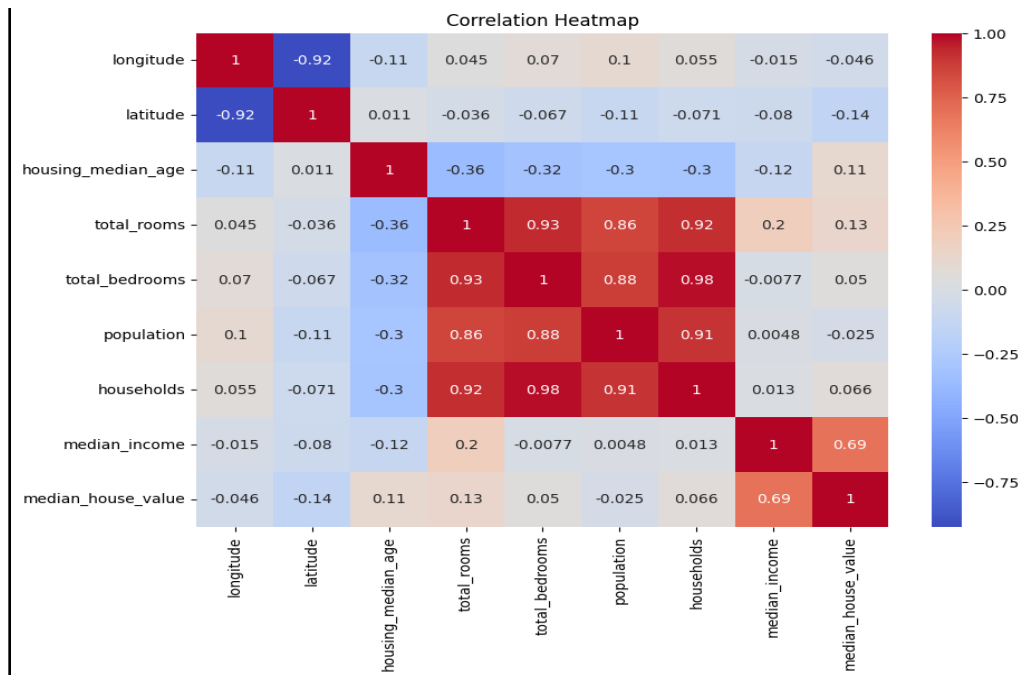
```
# Calculate Correlation Matrix

correlation = df.corr(numeric_only=True)
print(correlation)
```

```
***
   longitude  longitude  latitude  housing_median_age  total_rooms \
longitude      1.000000  -0.924664  -0.108197  -0.044568  0.044568
latitude      -0.924664  1.000000  0.011173  -0.036100  -0.036100
housing_median_age -0.108197  0.011173  1.000000  -0.361262  -0.361262
total_rooms      0.044568 -0.036100  -0.361262  1.000000  1.000000
total_bedrooms   0.069608 -0.066983  -0.320451  0.930380  0.930380
population       0.099773 -0.108785  -0.296244  0.857126  0.857126
households       0.055310 -0.071035  -0.302916  0.918484  0.918484
median_income    -0.015176 -0.079809  -0.119034  0.190850  0.190850
median_house_value -0.045967 -0.144160  0.105623  0.134153  0.134153
```

```
total_bedrooms  population  households  median_income \
longitude      0.069608  0.099773  0.055310  -0.015176
latitude      -0.066983 -0.108785 -0.071035  -0.079809
housing_median_age -0.320451 -0.296244 -0.302916 -0.119034
total_rooms    0.930380  0.857126  0.918484  0.190850
total_bedrooms 1.000000  0.877747  0.979728  -0.007723
population     0.877747  1.000000  0.907222  0.004834
households     0.979728  0.907222  1.000000  0.013033
median_income  -0.007723  0.004834  0.013033  1.000000
median_house_value 0.049686 -0.024650  0.065843  0.688075

median_house_value
longitude      -0.045967
latitude      -0.144160
housing_median_age 0.105623
total_rooms    0.134153
total_bedrooms 0.049686
population     -0.024650
households     0.065843
```

Output:

- B. Apply regression analysis to identify the relationship between an independent variable (e.g., advertising expenditure) and a dependent variable (e.g., sales revenue).

Code:

```
# Import Libraries

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
# Load Dataset

df = pd.read_csv("housing.csv")

# Select Independent and Dependent Variables

X = df[["median_income"]]
y = df["median_house_value"]
```

```
# Split Dataset into Training and Testing Data

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Linear Regression Model

model = LinearRegression()
```

```
# Train the Model

model.fit(X_train, y_train)

# Display Coefficient and Intercept

print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)

Coefficient: 41933.84939381272
Intercept: 44459.72916907875
```

```
# Predict House Values
```

```
y_pred = model.predict(X_test)
```

```
# Display First 10 Predictions
```

```
print("Predicted Values:")
```

```
print(y_pred[:10])
```

```
Predicted Values:
```

```
[114958.91676996 150606.88213964 190393.71844449 285059.38345102  
200663.31816103 242165.24890609 257647.22610228 199229.18051176  
245893.1681172 384677.43607096]
```

```
# Calculate Mean Squared Error
```

```
mse = mean_squared_error(y_test, y_pred)
```

```
print("Mean Squared Error:", mse)
```

```
Mean Squared Error: 7091157771.76555
```

```
# Calculate R2 Score
```

```
r2 = r2_score(y_test, y_pred)
```

```
print("R2 Score:", r2)
```

```
R2 Score: 0.45885918903846656
```

```
# Scatter Plot with Regression Line
```

```
plt.figure(figsize=(8, 5))
```

```
sns.regplot(  
    x="median_income",  
    y="median_house_value",  
    data=df  
)
```

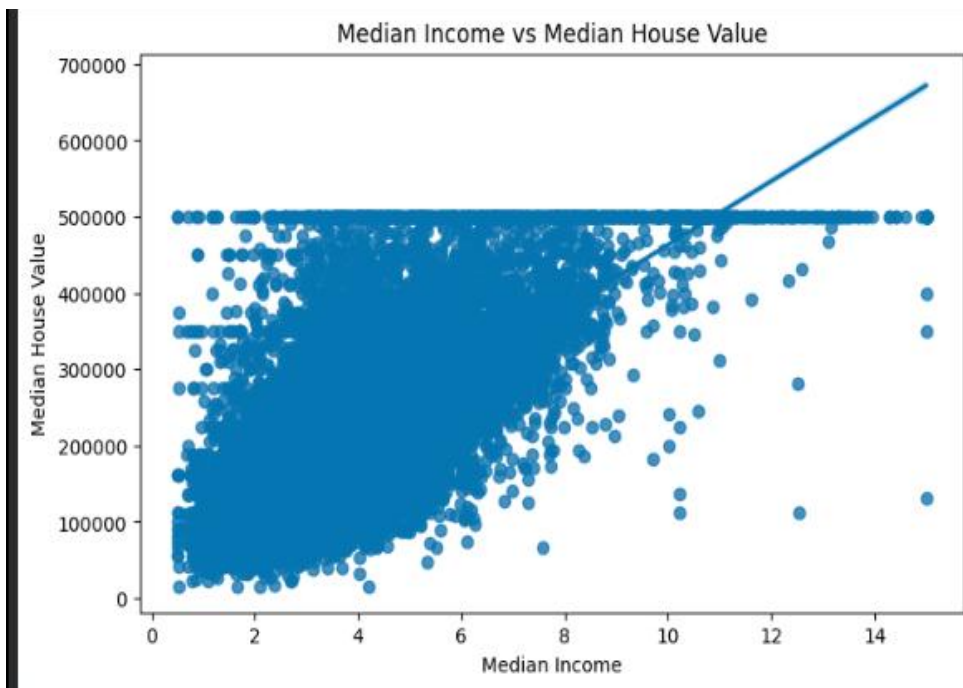
```
plt.xlabel("Median Income")
```

```
plt.ylabel("Median House Value")
```

```
plt.title("Median Income vs Median House Value")
```

```
plt.show()
```

Output:



Additional Questions on housing.csv (Must include in Print document)

Q.1 Calculate the complete correlation matrix for all numerical variables.

Identify: The strongest positive correlation, The strongest negative correlation, The weakest correlation.

Display the results.

Code:

```
import pandas as pd
import numpy as np
df = pd.read_csv("housing - housing.csv")
# Select numerical variables
numerical_data = df.select_dtypes(include=np.number)
# Calculate correlation matrix
correlation_matrix = numerical_data.corr()
print("Complete Correlation Matrix:")
print(correlation_matrix)
# Get correlation pairs
pairs = []
columns = correlation_matrix.columns
for i in range(len(columns)):
    for j in range(i + 1, len(columns)):
        pairs.append(
            (columns[i], columns[j], correlation_matrix.iloc[i, j])
        )
# Strongest positive correlation
strongest_positive = max(pairs, key=lambda x: x[2])
# Strongest negative correlation
strongest_negative = min(pairs, key=lambda x: x[2])
# Weakest correlation
weakest = min(pairs, key=lambda x: abs(x[2]))
print("\nStrongest Positive Correlation:")
print(strongest_positive)
print("\nStrongest Negative Correlation:")
print(strongest_negative)
print("\nWeakest Correlation:")
print(weakest)
```

Output:

```
Complete Correlation Matrix:
longitude  latitude  housing_median_age  total_rooms \
longitude    1.000000 -0.924664        -0.108197    0.044568
latitude    -0.924664  1.000000         0.011173   -0.036100
housing_median_age -0.108197  0.011173    1.000000   -0.361262
total_rooms    0.044568 -0.036100       -0.361262    1.000000
total_bedrooms  0.069608 -0.066983       -0.320451    0.930380
population     0.099773 -0.108785       -0.296244    0.857126
households     0.055310 -0.071035       -0.302916    0.918484
median_income  -0.015176 -0.079809       -0.119034    0.198050
median_house_value -0.045967 -0.144160    0.105623    0.134153

longitude  total_bedrooms  population  households  median_income \
longitude    0.069608    0.099773    0.055310   -0.015176
latitude    -0.066983   -0.108785   -0.071035   -0.079809
housing_median_age -0.320451 -0.296244 -0.302916 -0.119034
total_rooms    0.930380    0.857126    0.918484    0.198050
total_bedrooms  1.000000    0.877747    0.979728   -0.007723
population     0.877747    1.000000    0.907222    0.004834
households     0.979728    0.907222    1.000000    0.013033
median_income  -0.007723    0.004834    0.013033    1.000000
median_house_value  0.049686   -0.024650    0.065843    0.688075

longitude  median_house_value
longitude    -0.045967
...
('longitude', 'latitude', np.float64(-0.924664339150368))
```

Q.2 Build a regression model using: X=median_income, X=housing_median_age, X=total_rooms, X=population, X=households, Y=median_house_value

Evaluate the model using:

- R2
- Mean Squared Error

Compare this model with the simple regression model using only median_income.

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
df = pd.read_csv("housing - housing.csv")
# Independent variables
X = df[
    [
        "median_income",
        "housing_median_age",
        "total_rooms",
        "population",
        "households"
    ]
]
```

Output:

```
# Dependent variable
y = df["median_house_value"]
# Multiple Linear Regression
model = LinearRegression()
model.fit(X, y)
# Predictions
y_pred = model.predict(X)
# Evaluation
r2 = r2_score(y, y_pred)
mse = mean_squared_error(y, y_pred)
print("Multiple Linear Regression")
print("-----")
print("Intercept:", model.intercept_)
print("\nCoefficients:")
for feature, coefficient in zip(X.columns, model.coef_):
    print(feature, ":", coefficient)
print("\nR2 Score:", r2)
print("Mean Squared Error:", mse)
```

Multiple Linear Regression

Intercept: -37926.606155747344

Coefficients:

median_income : 45969.659381936
housing_median_age : 1842.1494875739568
total_rooms : -13.946204039687927
population : -39.44546486713395
households : 214.35386822082083

R2 Score: 0.5629054644856253

Mean Squared Error: 5820133599.282842

Q.3 Find the correlation between: total_rooms , median_house_value.

Code:

```
# Simple Linear Regression
X_simple = df[["median_income"]]
simple_model = LinearRegression()
simple_model.fit(X_simple, y)
# Predictions
y_pred_simple = simple_model.predict(X_simple)
# Evaluation
r2_simple = r2_score(y, y_pred_simple)
mse_simple = mean_squared_error(y, y_pred_simple)
print("Simple Linear Regression")
print("-----")
print("R2 Score:", r2_simple)
print("Mean Squared Error:", mse_simple)
```

Output:

Simple Linear Regression

R2 Score: 0.47344749180719903
Mean Squared Error: 7011311502.929527